

```
1. #!/bin/bash
echo "Enter file name:"
read filename
# Check if the file exists or not '-e'
if [-e "$filename"]; then
    echo "File '$filename' exists."
# Check if readable '-r'
    if [-r "$filename"]; then
        echo "File is readable"
    else
        echo "File is NOT readable"
    fi
# Check if writable '-w'
    if [-w "$filename"]; then
        echo "File is writable"
    else
        echo "File is not writable"
    fi
else
    echo "File '$filename' does not exist."
fi
```

```
1. SAY "Enter filename"
IF filename exists THEN
    SAY "Yes The file exists"
    IF filename can be read THEN
        SAY "You can read this file"
    ELSE
        SAY "You cannot read this file"
    END IF
ELSE
    IF FILENAME can be written THEN
        SAY "You can write this file."
    ELSE
        SAY "You cannot write to this file."
    END IF
ELSE
    SAY "File doesn't exist"
END IF
END
```

```
2. #!/bin/bash
echo "Enter the filename"
read filename

if [ -e "$filename" ]; then

#counts Lines line = $(wc -l < "$filename")
#count words words = $(wc -w < "$filename")
#count char char = $(wc -c < "$filename")

echo "File : $filename"
echo "lines : $line"
echo "words : $words"
echo "chars : $char"

else
echo "File '$filename' does not exist"
fi
```

```
SAY "Enter filename"
IF filename exists then
    count how many lines are in filename,
    call it line-count
    count how many words in filename,
    call it word-count
    count Number of characters in filename,
    call it char-count
```

```
SAY "This file has:"
SAY "line-count, 'lines'"
SAY "word-count, 'words'"
SAY "char-count, 'characters'"

ELSE
SAY "File doesn't exist"
ENDIF
END
```

3. #! /bin/bash

echo "Contents of current directory"

ls -l # lists all contents of the current directory

find the largest file & print its name and size

echo ""

echo "Largest file:"

ls -s -l | grep '^-' | head -1 | awk

{ print "Name:", \$NF, "Size:", \$5, "bytes" }

3.

START

SHOW ALL the files & folders in the current place

LOOK at every file in the current place
FIND the file that is the Biggest in size
~~and~~

REMEMBER its name, call it BIG-NAME

REMEMBER its size, call it BIG-SIZE

SAY "The biggest file ~~is~~:"

SAY "Name:", BIG-NAME

SAY "Size:", BIG-SIZE

END

4. #!/bin/bash

Ask user for a word

echo "Enter the word to search for"

read word

Ask the user for a filename

echo "Enter the filename:"

read filename

Check if the file exists

if [-e "\$filename"]; then

~~# Search~~ for the word and print matching lines
grep -n "\$word" "\$filename"

else

echo "File \$filename doesn't exist"

fi

START

SAY "Enter ~~file~~ word"

type something, call it word

SAY "Enter filename"

type something, call it File-name

If File-name exists THEN

LOOK At Every line in file name, one by one.

IF the line contains WORD THEN

SHOW that line

ELSE

SAY "Sorry, the file doesn't exist"

ENDIF

END

5. #!/bin/bash

~~#~~ Ask the user for a number

echo "Enter a number:"

read num

#Generate multiplication table from 1 to 20

i = 1

while [\$i -le 20]; do

result = \$((\$num * i))

echo "\$num x \$i = \$result"

i = \$((\$i + 1))

done

START

SAY "Enter a number"

+type some number, call it NUM

SET COUNTER to 1

WHILE COUNTER is less than or equal to 20

DO

MULTIPLY NUM by COUNTER, call it RESULT

SAY NUM, "x", COUNTER, "=", RESULT

ADD 1 to COUNTER

END WHILE

END

```
6:#!/bin/bash
# print current disk usage
echo "Current disk usage:"
df -h

# Check the usage percentage of the root filesystem
usage=$(df / | tail -1 | awk '{print $5}' | tr -d '%')
```

```
if ["$usage" -gt 80]; then
    echo "Warning!" $usage "%"
else
    echo "Disk usage is Normal $usage %"
fi
```

START

SHOW the current disk usage

FIND HOW full the disk is, call it USAGE
(a number like 85)

IF USAGE is greater than 80 THEN
SAY "WARNING", USAGE, "%"

ELSE

SAY "The disk is okay: ", USAGE, "%"

END IF

END

Lab-12

1.

```
#!/bin/bash
```

```
i=0
```

```
while [ $i -le 10 ]; do
```

```
    echo $i
```

```
    i=$((i+1))
```

```
done
```

1.

START

SET COUNTER to 0

WHILE COUNTER is less than or equal to 10 ; DO

SAY COUNTER

ADD 1 to COUNTER

END WHILE

END

2.

```
# /bin/bash
```

```
# print only even numbers from 0 to 50
```

```
for i in range $(seq 0 50); do
```

```
    if [  $$(($i \% 2))$  -eq 0 ]; then
```

```
        echo $i
```

```
    fi
```

```
done
```

or

```
# /bin/bash
```

```
for ((i=0; i<=50; i+=2)); do
```

```
    echo $i
```

```
done
```

2.

FOR EACH NUMBER from 0 to 50 DO

FIND the remainder when NUMBER is divided
by 2, call it ~~Remainder~~

If Remainder is equal to 0 THEN

SAY NUMBER

END IF

END FOR

END

3.

```
#!/bin/bash
```

```
# Ask the user for a number
```

```
echo "Enter a number:"
```

```
read num
```

```
# print multiplication tables upto 15
```

```
using until loop
```

```
i=1  
until [ $i -gt 15 ]; do
```

```
    result=$((num * i))
```

```
    echo "$num x $i = $result"
```

```
    i=$((i + 1))
```

```
done
```

3.

```
START
```

```
SAY "Enter a number"
```

```
WAIT for user to type something, call it NUM
```

```
SET COUNTER to 1
```

```
UNTIL COUNTER is greater than 15 DO
```

```
    MULTIPLY NUM by COUNTER, call it Result
```

```
    SAY NUM, "x", COUNTER, "=", Result
```

```
    ADD 1 to COUNTER
```

```
END UNTIL
```

```
END
```

4.

```
#!/bin/bash
```

```
echo "2 3 5 7" > num.txt
```

```
#read numbers from file into an array
```

```
read -ra number < num.txt
```

```
#print the elements of the array
```

```
echo "Original numbers:"
```

```
echo "${number[@]}"
```

```
#print doubled values using a for loop
```

```
echo "Doubled numbers:"
```

```
for num in "${number[@]}; do
```

```
    doubled=$((num*2))
```

```
    echo $doubled
```

```
done
```

4.

START

CREATE a file called "num.txt" with the these numbers written in it: 2 3 5 7

READ the numbers from "num.txt" into a list, call it NUMBERS

SAY "Original numbers:"

SHOW all the numbers in NUMBERS

SAY "Doubled numbers"

FOR EACH NUMBER in NUMBERS DO

MULTIPLY NUMBER by 2, call it DOUBLED

SAY DOUBLED

ENDFOR

END

5.

```
#!/bin/bash
```

```
divide() {
```

```
    local num1=$1
```

```
    local num2=$2
```

```
    local quotient
```

```
    local remainder
```

```
    if [ "$num2" -eq 0 ]; then
```

```
        echo "Error: Division by 0 not allowed"
```

```
        return 1
```

```
    fi
```

```
    quotient=$(echo "scale=2; $num1/$num2" | bc)
```

```
    remainder=$((num1 % num2))
```

```
    echo "Quotient: $quotient"
```

```
    echo "Remainder: $remainder"
```

```
}
```

```
echo "Enter first number: "
```

```
read a
```

```
echo "Enter Second number"
```

```
read b
```

```
divide "$a" "$b"
```

5.

```
START
```

```
Define function divide (NUM1, NUM2)
```

```
IF NUM2 equal to 0 THEN
```

```
    SAY "ERROR: Division by 0 not allowed"
```

```
    STOP the function here
```

```
END IF
```

Divide NUM1 by NUM2, keep 2 numbers after the dot, call it Quotient.

Divide NUM1 by NUM2, keep what is remaining, call it remainder

```
SAY "Quotient: quotient"
```

```
SAY "Remainder: remainder"
```

```
ENDFUNCTION
```

Ask user for two numbers, call the A and B

```
CALL divide (A,B)
```

```
END
```


6.

```
#!/bin/bash
maximum() {
    local num1=$1
    local num2=$2
    if [ "$num1" -gt "$num2" ]; then
        echo "Maximum is : $num1"
    else
        echo "Maximum is : $num2"
    fi
}
echo "Enter first number"
read a
echo "Enter first second number"
read b
maximum "$a" "$b"
```

6.

```
START
Define function maximum(NUM1, NUM2)
IF NUM1 is greater than NUM2 THEN
    SAY "Maximum is :", NUM1
ELSE
    SAY "Maximum is :", NUM2
ENDIF
END FUNCTION
```

```
ASK user for two number, call them A & B
CALL maximum (A, B)
END
```


7.

```
#!/bin/bash
```

```
check-directory() {  
    local dir=$1  
    if [ -d "$dir" ]; then  
        echo "Directory '$dir' exists, listing files"  
        ls -l "$dir"  
    else  
        echo "Directory $dir not exist. Creating new"  
        mkdir "$dir"  
        echo "Directory '$dir' Created"  
    fi  
}
```

```
echo "Enter directory name:"  
read dirname  
check-directory "$dirname"
```

7.

```
define function check-directory(DIR):
```

```
IF DIR exists THEN
```

```
    SAY "Directory exists, Listing files!"  
    SHOW ALL files in DIR
```

```
ELSE
```

```
    SAY "Directory not exist, Creating Directory"  
    MAKE a new directory called DIR  
    SAY "Directory created"
```

```
END IF
```

```
END function
```

```
ASK user for a directory name,  
CALL it NAME
```

```
CALL-Check-directory(NAME)  
END
```